WHY IT IS A GAME CHANGER

Close to 40 per cent of the 1.4 billion people in India do not have internet access, mostly in rural areas. Satellite internet is emerging as an effective alternative for communication and broadband services in remote areas where other traditional internet mediums like DSL and cable are not available. Demand for instant communication and increased smartphone penetration has led to big internet companies entering this space to offer faster internet networks to consumers. Analysts say the satellite internet market has the potential to generate $1.9 billion (Rs 15,800 crore) in revenue in India by 2030.

The state-owned Bharat Broadband Network Ltd (BBNL) is implementing the BharatNet project—the rollout of satellite internet services to remote regions—and has finished pilot tests in Arunachal Pradesh. It will connect 7,000 gram panchayats across India. BBNL is already providing internet services—serviced by Indian Space Research Organisation (ISRO) satellites—to soldiers deployed at the Siachen glacier, enabling them to talk to their families.

WHAT INDIA NEEDS TO MASTER

The entry of new and serious players is what has led to the rapid pace of digitisation and innovations in the satellite and space industry. Industry experts say a global transformation is expected in the coming years. Satellite and space are fixed and high-cost industries, hence only market consolidation or partnerships can ensure financial benefits and constant capital flow. High production costs for launching satellites and installing ground equipment are ultimately passed on to the end-users. The challenge is to reduce these costs. India is at an advantage as ISRO offers launches at a fraction of the cost charged by other space-faring nations. After a regulatory overhaul, India should become a favoured low-cost destination for satellite launches.

Further, satellite internet has the potential to trigger disruptions in the telecom market. The increasing number of satellites in orbit will help satellite operators grow and expand service offerings. Satellite providers and telecom majors joining hands can, along with current offerings, also provide a voice-based telephone service for users to make satellite-based calls. ■

THE CHANGE-MAKERS

ONE WEB

- A global communications network backed by the UK government with Bharti Global as a strategic stakeholder, it is powered by a constellation of 634 low Earth orbit (LEO) satellites that are delivering high speed, low latency connectivity to even the hardest-to-reach places. In May, it completed the 19th launch
- Has partnered with Hughes Communications India Pvt Ltd (HCIPL) to bring connectivity to towns, villages and businesses across India
- Aviation connectivity is being transformed with broadband-in-the-sky connectivity. OneWeb has also signed an MoU with Gujarat's science and technology department in July to set up a satellite network portal site in Mehsana

BHARAT BROADBAND NETWORK LTD

- The BharatNet project of the state-backed BBNL aims to connect 7,000 gram panchayats across India through satellite internet

JIO PLATFORMS

- The Reliance subsidiary is planning a next-gen scalable and affordable broadband services company in a joint venture with Luxembourg-based satellite services provider SES

STARLINK

- The satellite internet constellation operated by Elon Musk's SpaceX has applied to establish earth stations in India



"The scale and adaptability of low Earth orbit (LEO) satellite networks creates a degree of future-proofing that will keep pace with demands for bandwidth and speed. The deployment of our Gen-2 satellite fleet will help in this"

–MASSIMILANO LADOVAZ
CTO, OneWeb